

Lecture 5

Automation

Empirical Macroeconomics

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Industrial Robots and Automation



Industrial Robots

- Automatic and multipurpose machines with several axis suitable for industrial applications.
- Tasks performed by industrial robots used to be labor intensive (machining, welding, assembling, inspecting, packaging).
- Large increase between 1993 and 2014:
 - In the US, fivefold increase from 2 to 10 robots per thousand industry workers.
 - Even more in other countries (e.g. Germany, Japan)

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- Tasks performed by industrial robots used to be labor intensive (machining, welding, assembling, inspecting, packaging).
- Large increase between 1993 and 2014:
 - In the US, fivefold increase from 2 to 10 robots per thousand industry workers.
 - Even more in other countries (e.g. Germany, Japan)
- Industrial robots are leading example of an **automation technology**
 - Other automation technologies? Perhaps artificial intelligence (AI)
 - Today's question: *what is the effect of automation on labor markets?*

Acemoglu & Restrepo (2020)

1. Model to clarify impact of automation

- In each industry, robots compete task-by-task against labor.
- Automation: differential increase in the share of tasks that robots can perform in some industries.
- Automation raises **productivity**, but also **displaces labor**.
- In each labor market, net effect of robots summarized by:

$$\text{Exposure to robots in } c = \sum_i \text{Base employment in industry } i \text{ at } c \times \text{National increase in robots per worker in } i$$

- In the presence of trade: local elasticity of robots $\neq$ aggregate elasticity of robots.

Acemoglu & Restrepo (2020)

2. Estimate impact of robots in US counties between 1990-2007

- Compute exposure to robots for counties using data on the use of robots by industry.
- Instrumented using adoption in 9 European countries. (as in Autor, Dorn and Hanson, 2013; and Bloom, Draca and Van Reenen, 2015).
- Negative impact of robots on employment and wages.
- Using shift share instrument derived from model

Acemoglu & Restrepo (2020)

3. Quantitative implications

- Use model to compute aggregate effect accounting for trade.
- Effect since 1990: reduction of 660,000 jobs and 0.5% in wages.

Today's Plan

Discuss Acemoglu & Restrepo (2020) at length:

- Well executed and influential paper (5100+ citations already!)
- Important question
- Great use of model for multiple purposes
- High quality empirical work: even without “ideal natural experiment”

Also:

- Chance to digress into shift share instruments

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A Model of Automation

- Commuting zones $c \in \mathcal{C}$.
- Final good Y_c combining the output of industries $i \in \mathcal{I}$:

$$Y_c = \left(\sum_{i \in \mathcal{I}} \alpha_i^{\frac{1}{\sigma}} Y_{ci}^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}$$

where α_i is a share parameter.

- **For now—no trade:** industry output produced locally

A Model of Automation

- Labor L_c supplied in each czone at wage W_c :

$$W_c = \Phi Y_c L_c^\varepsilon.$$

Frisch elasticity $1/\varepsilon$ and negative income effect Y_c .

- Reallocation within czones but not across.
- Robots supplied with a price-elasticity η :

$$R_c = \Psi Q_c^\eta$$

$\frac{\eta}{1+\eta}$ = share of factors complementary to robots that earn rents.

Tasks and Automation (1/2)

- Each industry combines tasks $x_{ci}(s)$ from a continuum $s \in [0, 1]$.
- These tasks must be combined in fixed proportions:

$$X_{ci} = A_{ci} \min_{s \in [0, 1]} \{x_{ci}(s)\}.$$

- A_{ci} drives comparative advantage of regions in different industries.

Tasks and Automation (1/2)

- In industry i tasks $[0, M_i]$ can be performed by robots, $q_{ci}(s)$; while the remaining tasks must be produced with labor, $l_{ci}(s)$:

$$x_{ci}(s) = \begin{cases} \gamma q_{ci}(s) + l_{ci}(s) & \text{if } s \leq M_i \\ l_{ci}(s) & \text{if } s > M_i, \end{cases}$$

- “Constant” input requirements: $\frac{M_i}{A_{ci}\gamma}$ robots and $\frac{1-M_i}{A_{ci}}$ units of labor.
- **Automation:** Technological improvements that raise M_i .
- $dM_i \geq 0$ varies by industry and reduces the labor share, s_{ci}^L .

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- “Constant” input requirements: $\frac{M_i}{A_{ci}\gamma}$ robots and $\frac{1-M_i}{A_{ci}}$ units of labor.
- **Automation:** Technological improvements that raise M_i .
- $dM_i \geq 0$ varies by industry and reduces the labor share, s_{ci}^L .
- Discuss: how is this model different from the “canonical” production function in capital and labor?

Partial Equilibrium: The Demand for Labor

Proposition. Let P_{Xci} denote the price of Y_{ci} . The total demand for labor L_c^d in commuting zone c satisfies:

$$d \ln L_c^d = - \sum_{i \in \mathcal{I}} \ell_{ci} \frac{dM_i}{1 - M_i} - \sigma \sum_{i \in \mathcal{I}} \ell_{ci} d \ln P_{Xci} + d \ln Y_c,$$

where ℓ_{ci} is the share of employment in industry i and czone c .

- The first term is the **displacement effect**
- The second term is the **price-productivity effect** → depends on π_c .
- The third term is the **scale-productivity effect** → depends on π_c .

(π_c is the cost savings from adopting robots)

General Equilibrium Impact and Main Equation

Assumption. Let $\pi_c = 1 - \frac{R_c}{\gamma W_c}$ denote the cost-saving gains from using robots. For all commuting zones $c \in \mathcal{C}$ we have $\pi_c > 0$.

→ All tasks $i < M_i$ are automated, remaining tasks done with labor

Proposition. Assume initially $M_i = M$. In autarky, the impact of robots on **total employment** and **wages** is given by

$$d \ln L_c = \left(-\frac{1 + \eta}{1 + \varepsilon} + \frac{1 + \eta}{1 + \varepsilon} \pi_c \right) \sum_{i \in \mathcal{I}} \ell_{ci} \frac{dM_i}{1 - M_i}$$
$$d \ln W_c = (-\eta + (1 + \eta) \pi_c) \sum_{i \in \mathcal{I}} \ell_{ci} \frac{dM_i}{1 - M_i},$$

where ℓ_{ci} is the share of employment in industry i and czone c .

Discussion

- Automation could reduce employment and wages.
 - Due to “displacement effect” of robots replacing workers in tasks
- But factor-neutral productivity gains or increase in generic capital won't
 - Increase in generic capital **always** raise labor demand
 - In $Y = F(K, L)$, K and L are **q-complements**

Incorporating Trade

- X_{ci} is now a differentiated good traded across commuting zones (as in Armington, 1969), and:

$$Y_{ci} = \left(\sum_{s \in \mathcal{C}} \theta_{si} X_{sci}^{\frac{\lambda-1}{\lambda}} \right)^{\frac{\lambda}{\lambda-1}},$$

with X_{sci} the amount of good i exported from s to c .

- Because commuting zones trade, we now have that

$$X_{ci} = \sum_{d \in \mathcal{C}} X_{cdi},$$

and X_{cdi} is sold at a price P_{Xci} .

- We assume there are no trade costs, so that productivity gains are shared uniformly.

Equilibrium Impact With Trade

Proposition. Assume initially $M_i = M$ and $\pi_c = \pi$. The **local effect** of robots on employment and wages is given by

$$d \ln L_c = (-\Phi_d^L + \Phi_\pi^L \pi) \sum_{i \in \mathcal{I}} \ell_{ci} \frac{dM_i}{1 - M_i} + \Theta_n^L$$

$$d \ln W_c = (-\Phi_d^W + \Phi_\pi^W \pi) \sum_{i \in \mathcal{I}} \ell_{ci} \frac{dM_i}{1 - M_i} + \Theta_n^W,$$

where the Φ 's are functions of the elasticities $\{\lambda, \sigma, \eta, \varepsilon\}$ and factor shares, and the Θ 's depend on automation in all czones. **The aggregate effects** are:

$$d \ln L_{US} = \left(-\frac{1 + \eta}{1 + \varepsilon} + \frac{1 + \eta}{1 + \varepsilon} \pi \right) \sum_{i \in \mathcal{I}} \ell_i \frac{dM_i}{1 - M_i}$$

$$d \ln W_{US} = (-\eta + (1 + \eta)\pi) \sum_{i \in \mathcal{I}} \ell_i \frac{dM_i}{1 - M_i},$$

Uses of a Model

✓ Conceptual points about automation:

- Automation can reduce labor demand
- Unlike generic forms of capital

✓ Motivating regression specification:

$$d \ln L_c = \left(-\phi_d^L + \phi_\pi^L \pi \right) \sum_{i \in \mathcal{I}} \ell_{ci} \frac{dM_i}{1 - M_i} + \Theta_n^L$$

looks just like a regression equation:

- Shift share instrument
- Time fixed effects

✓ Explaining difference between regional and aggregate objects

✓ Aggregate quantification

- Can easily be done with regression estimates + simple calibration

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Empirical Specification

- The term $\sum_{i \in \mathcal{I}} \ell_{ci} \frac{dM_i}{1-M_i}$ summarizes the impact of automation
- For $M_i \approx 0$, we have

$$\frac{dM_i}{1-M_i} = \gamma \frac{dQ_i}{L_i}.$$

- Long differences specification over 1990-2007 for 722 US czones:

$$\Delta \ln L_c = \gamma \left(-\phi_d^L + \phi_\pi^L \pi \right) \sum_{i \in \mathcal{I}} \ell_{ci}^{1990} \frac{\Delta Q_i^{US}}{L_i} + \epsilon_c^L$$

$$\Delta \ln W_c = \gamma \left(-\phi_d^W + \phi_\pi^W \pi \right) \sum_{i \in \mathcal{I}} \ell_{ci}^{1990} \frac{\Delta Q_i^{US}}{L_i} + \epsilon_c^W.$$

- ℓ_{ci}^{1990} : is the czone share of employment in each industry in 1990, ΔQ_i^{US} is data from the IFR on the use of robots
 - (Three-digit manufacturing industries and two-digit nonmanufacturing industries)

Omitted Variable Bias and IV Strategy

- Two issues with measure of exposure:
 1. Model assumes that robot adoption driven only by exogenous technological changes (Not omitted variables e.g. local product demand shocks)
 2. Measurement error
- To address both concerns, instrument with predicted exposure:

$$\text{predicted exposure to robots, 1993-2007}_c = \sum_{i \in \mathcal{I}} \ell_{ci}^{1970} \frac{\Delta Q_{i,1993-2007}^{\text{Euro}}}{L_i},$$

using data 1993-2007 for European countries (**shift share instrument**)

- Rationale:
 1. Europe (and Japan) world leaders in robotics, US adopts technologies from them.
 2. Employment shares in 1970 capture persistent differences in industry composition.

Adoption of Robots Over Time

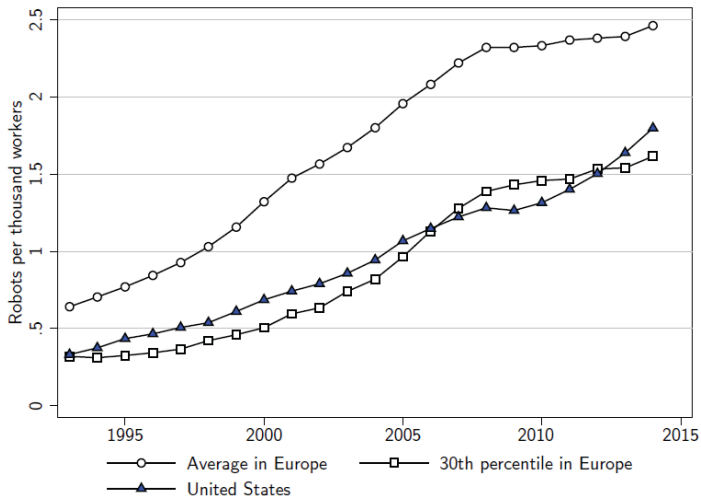


Figure: Trends in the use of industrial robots per thousand workers. Data: IFR.

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Shift-Share Instruments Primer

- Acemoglu & Restrepo are using a shift share instrument
- Very popular way to generate “reasonably exogenous” variation
 - When there is no natural experiment
- Digression to explain shift share instruments
 - Draws on Peter Hull’s excellent slides

Introduction to Shift Share Instruments

- Shift share instruments:
 - Bartik (1991): predicted employment growth $z_\ell = \sum_n s_{\ell n} g_n$ as an IV for region ℓ 's employment growth, where g_n is the national growth of industry n and the $s_n \in [0, 1]$ are lagged employment shares
 - Similar instruments in Blanchard & Katz (1992), Card (2009), Autor, Dorn & Hanson (2013), **Acemoglu & Restrepo (2020)**
- Recent literature: how “shift-share” IVs can be used for causal inference
 - Formalizes two paths to identification: via “shocks” g_n or “shares” s_n
 - Raises new practical considerations for SSIV estimation and inference

The SSIV Setting

- Suppose we are interested in estimating some parameter β of a linear causal or structural model $y_\ell = \beta x_\ell + e_\ell$
- Residualize e_ℓ on a vector of observed controls w_ℓ to get second stage:

$$y_\ell = \beta x_\ell + w_\ell' \gamma + \varepsilon_\ell$$

where w_ℓ and ε_ℓ are orthogonal by construction: $E[\sum_\ell w_\ell \varepsilon_\ell] = 0$

- We instrument x_ℓ with $z_\ell = \sum_n s_{\ell n} g_n$, where $\sum_n s_{\ell n} = 1$
 - Call $s_{\ell n}$ the “exposure shares” and g_n the “shocks”
 - Share vary across observations, shocks do not
- IV is valid if $E \left[\frac{1}{L} \sum_\ell z_\ell \varepsilon_\ell \right] = 0$; identification follows from a first stage

The SSIV Estimator

- SSIV divides the regression of y_ℓ on z_ℓ , controlling for w_ℓ , (“reduced form”) by the regression of x_ℓ on z_ℓ , controlling for w_ℓ (“first stage”)
- By the Frisch-Waugh-Lovell theorem, this estimator can be written

$$\hat{\beta} = \frac{\sum_{\ell} z_{\ell} y_{\ell}^{\perp}}{\sum_{\ell} z_{\ell} x_{\ell}^{\perp}} = \frac{\sum_{\ell} \sum_n s_{\ell n} g_n y_{\ell}^{\perp}}{\sum_{\ell} \sum_n s_{\ell n} g_n x_{\ell}^{\perp}}$$

where $v_{\ell}^{\perp}$ denotes sample residuals from regressing v_{ℓ} on w_{ℓ}

- Plugging in the model $y_{\ell} = \beta x_{\ell} + w'_{\ell} \gamma + \varepsilon_{\ell}$ gives

$$\hat{\beta} = \beta + \frac{\sum_{\ell} \sum_n s_{\ell n} g_n \varepsilon_{\ell}^{\perp}}{\sum_{\ell} \sum_n s_{\ell n} g_n x_{\ell}^{\perp}}.$$

- Standard notions of consistency and inference

Challenges

1. Identification from Shares: Goldsmith-Pinkham, Sorkin & Swift (2020)
2. Identification from Shocks: Borusyak, Jaravel & Hull (2020)
 - Also has a good resolution to some novel inference challenges

Goldsmith-Pinkham, Sorkin, and Swift (GPSS; 2020)

- GPSS are interested in understanding when/how SSIV can be seen as leveraging quasi-experimental variation across observations
 - Viewing the g_n as fixed, $z_\ell = \sum_n s_{\ell n} g_n$ is a linear combination of shares
 - It follows that z_ℓ is a valid instrument when the shares are exogenous
- Formally, GPSS establish a numerical equivalence:
 - $\hat{\beta}$ can be obtained from an overidentified IV procedure that uses N share instruments $s_{\ell n}$ and a weight matrix based on the shocks g_n
- In the second stage regression

$$y_\ell = \beta x_\ell + w_\ell' \gamma + \varepsilon_\ell$$

instrument with $s_{\ell 1}, \dots, s_{\ell N}$ using fixed coefficients
(instead of estimated coefficients as is standard)

Goldsmith-Pinkham, Sorkin, and Swift (GPSS; 2020)

- Sufficient condition for identification: quasi-experimental assignment of shares across observations

$$E[\varepsilon_\ell | s_{\ell n}] = 0, \forall n \implies E\left[\frac{1}{L} \sum_{\ell} z_{\ell} \varepsilon_{\ell}\right] = \frac{1}{L} \sum_{\ell} \sum_n g_n E[s_{\ell n} E[\varepsilon_{\ell} | s_{\ell n}]] = 0$$

- Standard requirement that instruments are randomly assigned
- Diff-in-diff logic: suppose ε_{ℓ} are unobserved outcome trends
 - As in Acemoglu/Restrepo
 - $E[\varepsilon_{\ell} | s_{\ell n}] = 0$ is akin to a “parallel trends” assumption
 - Areas with different industry composition evolve similarly absent automation
 - Does not seem very realistic—will revisit

Rotemberg Weights

- The GPSS view of SSIV is one of many share instruments
 - In Acemoglu/Restrepo there are 19 industries = 19 instruments
- They propose “opening the black box” of overidentified IV by deriving the weights SSIV implicitly puts on each share instrument
 - Builds on Rotemberg (1983), so they call these “Rotemberg weights”

$$\hat{\beta} = \sum_n \hat{\alpha}_n \hat{\beta}_n \quad \text{where} \quad \underbrace{\hat{\beta}_n = \frac{\sum_{\ell} s_{\ell n} y_{\ell}^{\perp}}{\sum_{\ell} s_{\ell n} x_{\ell}^{\perp}}}_{\text{Industry specific estimate}} \quad \text{and} \quad \hat{\alpha}_n = \frac{g_n \sum_{\ell} s_{\ell n} x_{\ell}^{\perp}}{\sum_{n'} g_{n'} \sum_{\ell} s_{\ell n'} x_{\ell}^{\perp}}$$

- Intuitively, more weight is given to share instruments with higher shocks g_n and larger first stages $\sum_{\ell} s_{\ell n} x_{\ell}^{\perp}$

Rotemberg Weights in Acemoglu / Restrepo

	WEIGHTED BY POPULATION				UNWEIGHTED	
	RAW DATA	BASELINE COVARIATES	EXCLUDES ZONES WITH HIGHEST EXPOSURE	EXCLUDES AUTOMOTIVE MANUFACTURING	BASELINE COVARIATES	EXCLUDES AUTOMOTIVE MANUFACTURING
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A. Long differences, 1990-2007</i>						
Metal products	.035	-.002	.035	.046	.029	.055
Basic Metals	.037	-.018	.008	.03	.017	.035
Plastics and Chemicals	.158	.193	.565	.949	.437	.896
Automotive	.776	.877	.417	-.001	.542	0
<i>Panel B. Stacked differences, 1990-2000 and 2000-2007</i>						
Food and Beverages 00-07	-.006	-.008	-.002	.002	0	.019
Wood and Furniture 00-07	-.005	-.008	-.011	-.004	-.003	.017
Industrial Machinery 90-00	-.002	.005	.01	.01	.007	.008
Mining 90-00	0	.005	.01	.013	.011	.022
Shipbuilding and Aerospace 90-00	.002	.001	.012	.024	.004	.015
Electronics 00-07	.013	.005	.062	.068	.032	.05
Industrial Machinery 00-07	.016	.01	.026	.023	.015	.017
Automotive 90-00	.018	-.035	-.028	0	-.018	-.001
Basic Metals 00-07	.019	.01	.036	.045	.025	.032
Metal products 00-07	.032	.025	.071	.085	.053	.085
Plastics and Chemicals 90-00	.049	.034	.104	.346	.096	.381
Plastics and Chemicals 00-07	.059	.059	.184	.393	.154	.362
Automotive 00-07	.795	.906	.546	-.001	.644	-.001

Is Share Exogeneity a Plausible Identifying Assumption?

- Several ways to probe the plausibility of exogenous $s_{\ell n}$ ex post:
 - Balance/pre-trend tests
 - Straightforward to implement; no different than any other IV
 - Acemoglu / Restrepo pass these tests
- In some settings, share exogeneity is *ex ante* implausible
 - Areas with different industry composition probably have different outcomes for lots of different reasons!
 - Would you use industry shares as an instrument for *anything*?
- Likely share endogeneity calls for a new approach to SSIV

Challenges

1. Identification from Shares: Goldsmith-Pinkham, Sorkin & Swift (2020)
2. **Identification from Shocks: Borusyak, Jaravel & Hull (2020)**
 - Also has a good resolution to some novel inference challenges

Borusyak, Hull, and Jaravel (BHJ; 2020)

- BHJ are interested in understanding when/how SSIV can be seen as leveraging quasi-random variation in the shocks
- Like GPSS, they establish a numerical equivalence:
 - $\hat{\beta}$ can be obtained from a just-identified shock-level IV procedure that uses g_n to instrument for a shock-level “aggregate” of the treatment:

$$\hat{\beta} = \frac{\sum_{\ell} z_{\ell} y_{\ell}^{\perp}}{\sum_{\ell} z_{\ell} x_{\ell}^{\perp}} = \frac{\sum_{\ell} \sum_n s_{\ell n} g_n y_{\ell}^{\perp}}{\sum_{\ell} \sum_n s_{\ell n} g_n x_{\ell}^{\perp}} = \frac{\sum_n g_n \sum_{\ell} s_{\ell n} y_{\ell}^{\perp}}{\sum_n g_n \sum_{\ell} s_{\ell n} x_{\ell}^{\perp}} = \frac{\sum_n g_n \bar{y}_{\ell}^{\perp}}{\sum_n g_n \bar{x}_{\ell}^{\perp}}$$

where $s_n = \frac{1}{L} \sum_{\ell} s_{\ell n}$ are weights capturing the average importance of shock n and $\bar{v}_n = \frac{\sum_{\ell} s_{\ell n} v_{\ell}}{\sum_{\ell} s_{\ell n}}$ is an exposure weighted average of v_{ℓ}

BHJ as Weighted Regression

- Can estimate

$$\hat{\beta} = \frac{\sum_n g_n \bar{y}_\ell^\perp}{\sum_n g_n \bar{x}_\ell^\perp} \quad \bar{v}_n = \frac{\sum_\ell s_{\ell n} v_\ell}{\sum_\ell s_{\ell n}}$$

from a **weighted shock level IV regression**

$$\bar{y}_n^\perp = \alpha + \beta \bar{x}_n^\perp + q_n' \tau + \bar{\varepsilon}_n^\perp$$

- Example from Acemoglu / Restrepo:
 - Industry regression with appropriate (but unusual) weights
- They then derive new conditions for SSIV identification + inference

BHJ Baseline Identification Assumptions

- **A1** (Quasi-random shock assignment): $E[g_n|\bar{\varepsilon}, s] = \mu, \forall n$
 - Each shock has the same expected value, conditional on the shock-level unobservables $\bar{\varepsilon}_n$ and average exposure s_n
 - Implies SSIV validity: $E\left[\frac{1}{L} \sum_{\ell} z_{\ell} \varepsilon_{\ell}\right] = E\left[\sum_n s_n g_n \bar{\varepsilon}_n\right] = 0$
- **A2** Many uncorrelated shocks for asymptotics
- Example from Acemoglu / Restrepo:
 - **A1**: shocks to industry automation uncorrelated with other factors (Remember use of foreign automation penetration in instrument)
 - **A2**: need sufficiently many industries

Ex Ante vs. Ex Post Validity

- BHJ emphasize that the decision to pursue a “shocks” vs. “shares” identification strategy should be made ex ante
 - Undesirable to base identifying assumptions on ex post tests, though balance/pre-trend tests can be used to falsify assumptions
 - The two identification strategies may have different economic content
- They suggest thinking about whether shares are “tailored” to the economic question / treatment, or are “generic”
 - Generic shares (e.g. Acemoglu/Restrepo): unobserved shocks likely to affect industry shares, violating share exogeneity
 - Tailored shares have a difference-in-difference feel; don't even need the shocks, except to possibly improve power / avoid many-IV bias

Shift Share Inference

- Adao, Kolesar & Morales (2020) study a novel inference challenge when SSIV identification leverages quasi-random shocks
 - Observations with similar shares $s_{\ell 1}, \dots, s_{\ell N}$ are likely to have correlated z_{ℓ} , even when not “clustered” in conventional ways (e.g. by distance)
 - When $\bar{\varepsilon}$ is similarly clustered (e.g. $\varepsilon_{\ell} = \sum_n s_{\ell n} \nu_n + \tilde{\varepsilon}_{\ell}$), the large-sample distribution of $\hat{\beta}$ may not be well-approximated by standard CLTs
- They show by simulation that this can lead to large size distortions
 - Tests with nominal 5% rejection rates can reject true nulls in 55% of placebo shock realizations (ADH-based Monte Carlo)
 - Reminiscent of Bertrand et al. (2004) study of robust SEs in diff-in-diff

Exposure-Robust SEs

- BHJ show a convenient solution to exposure clustering, via their equivalent shock-level IV regression
- Usual robust/clustered SEs can be valid when $\hat{\beta}$ is given by estimating

$$\bar{y}_n^\perp = \alpha + \beta \bar{x}_n^\perp + q_n' \tau + \bar{\varepsilon}_n^\perp$$

instrumenting $\bar{x}_n^\perp$ by g_n and weighting by s_n

- Clustering logic: can get valid SEs by estimating the IV at the level of the identifying variation (here, shocks)

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Reduced-form evidence

- Long-differences specification over 1990-2007 for 722 US czones:

$$\Delta \ln Y_c = \gamma \cdot \begin{array}{c} \text{exogenous exposure} \\ \text{to robots from 1993 to 2007}_c \end{array} + v_c$$

- Exogenous change in exposure to robots among European industries:

$$\begin{array}{c} \text{exogenous exposure} \\ \text{to robots, 1993-2007}_c \end{array} = \sum_{i \in \mathcal{I}} \ell_{ci} \left[p_{30} \left(\frac{R_{i,2007}}{L_i} \right) - p_{30} \left(\frac{R_{i,1993}}{L_i} \right) \right].$$

- p_{30} = 30th percentile of robot adoption among 9 European countries.
- IFR data on the use of robots in 3-digit manufacturing industries and 2-digit nonmanufacturing industries.
- ℓ_{ci} is the czone share of employment in each industry (Census data for 722 commuting zones in the US)
- Change in employment and wage data: Census and American Community Survey.

Exposed Industries: Europe and US Adoption

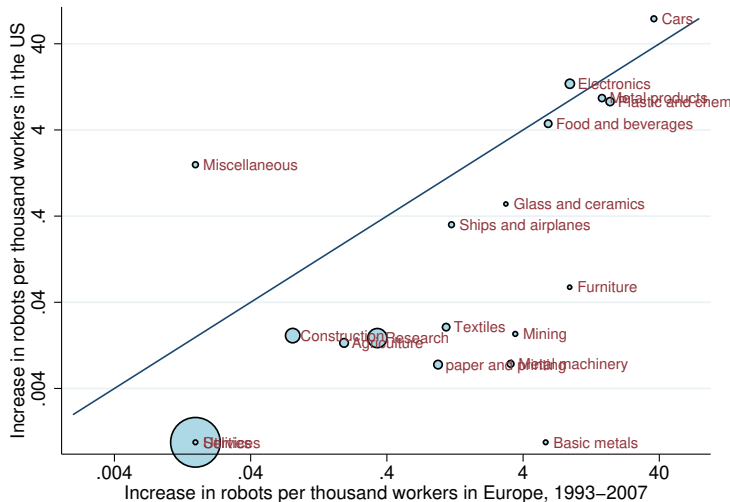


Figure: Increase use of robots in US and European industries. Data from the IFR. Both axis 43 / 73

Exposed Industries: Labor Share

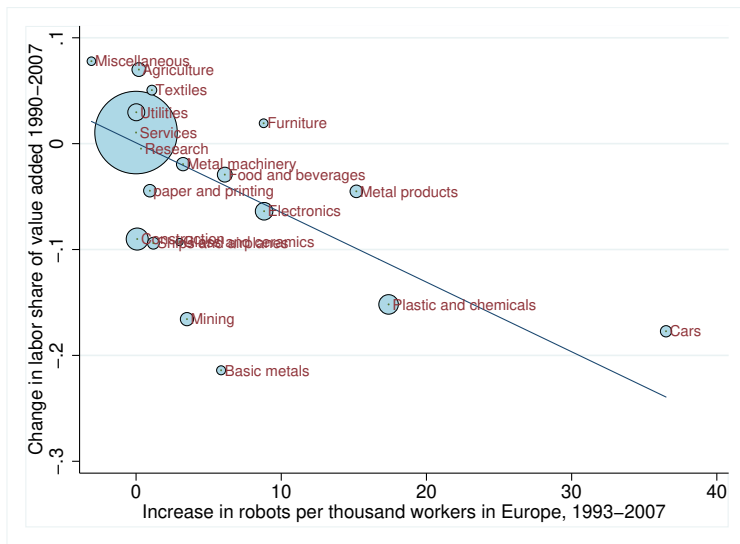


Figure: Change in the labor share and predicted increase in the use of robots among

Exposed Industries: Production Workers

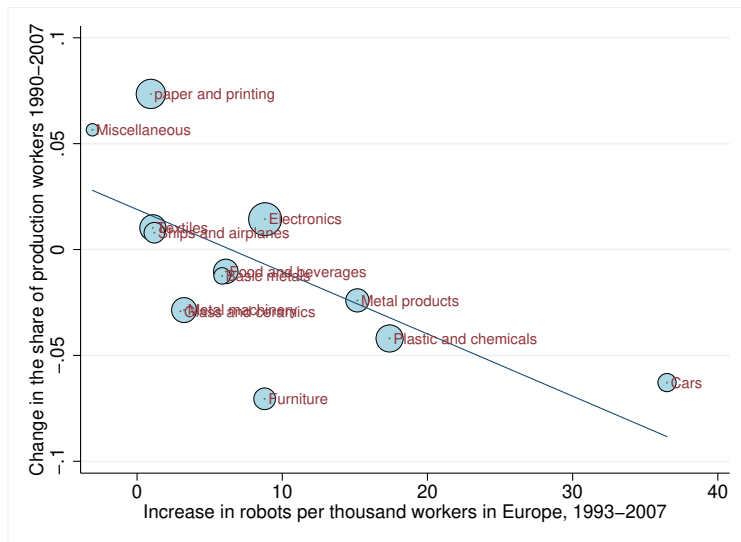


Figure: Change in the share of production workers and predicted increase in the use of

Exposed Industries: Output

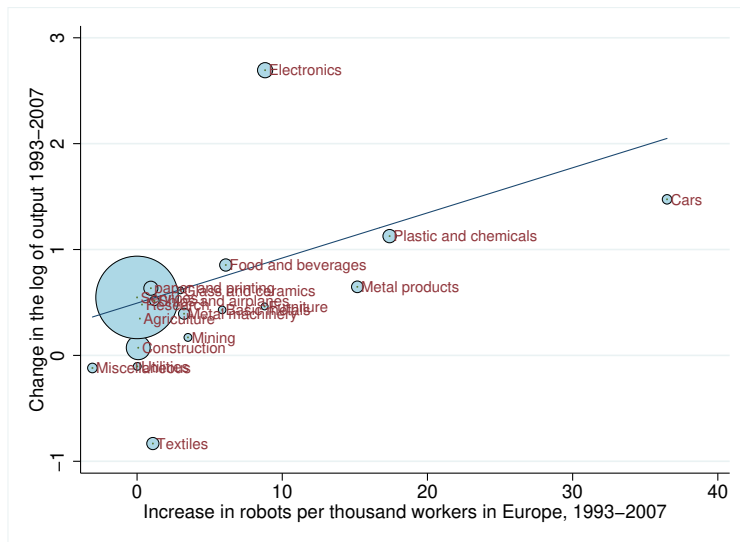


Figure: Change in the log of quantities produced (index) and predicted increase in the use

Exposed Industries: Employment

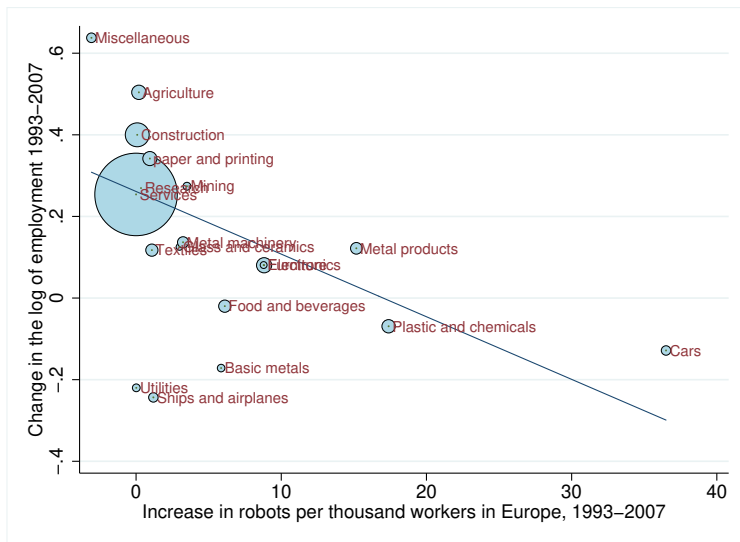


Figure: Change in the log of employment and predicted increase in the use of robots among 47 / 73

Exposed Industries—Other Shocks

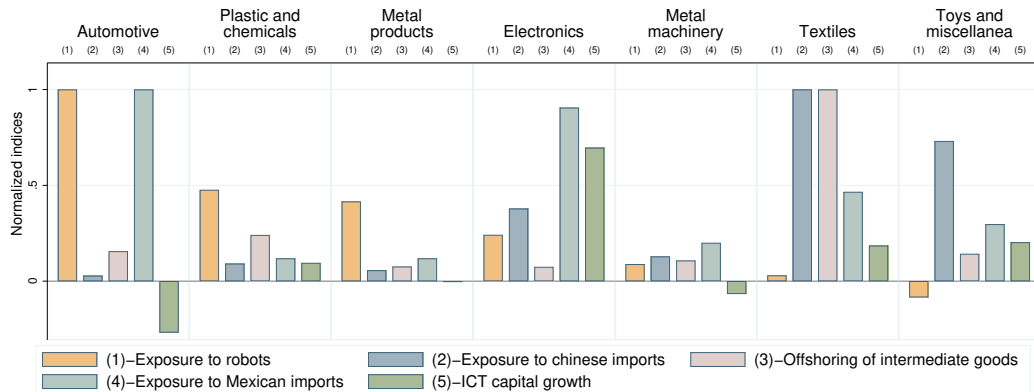


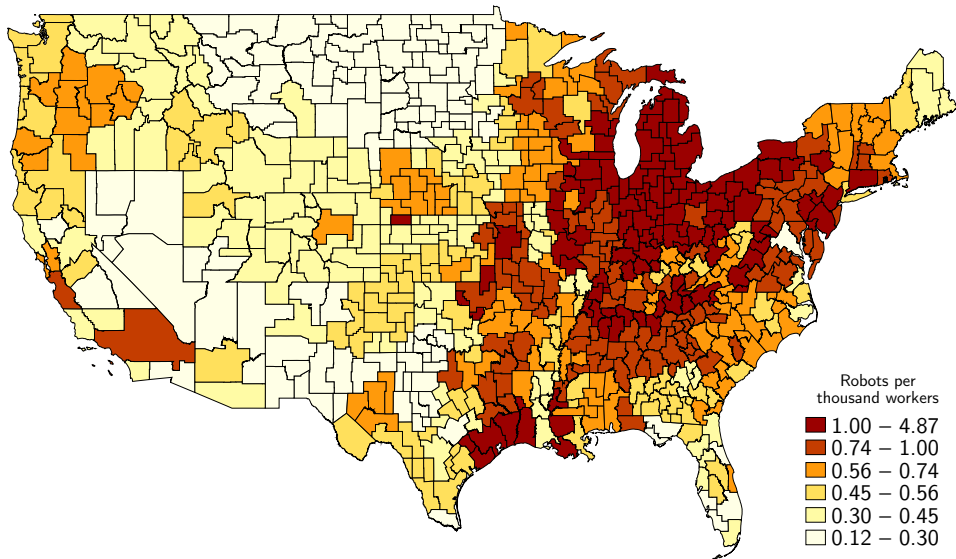
Figure: Increase in robots and other shocks affecting selected US industries.

Exposed Industries: Summary

- Adoption of robots driven by worldwide technological improvements that make robots useful in more applications.
 1. US adopting robots in industries where Europeans are making mayor inroads in development.
 2. Engineering literature supports this view.
- Unlike other forms of capital and technologies, the adoption of robots seems closely linked to the displacement of labor from remaining labor-intensive tasks in manufacturing.
- As a result, industries with greatest use of robots experiencing a decline in their labor share and demand for production workers.
- Manufacturing became less labor intensive, particularly in some industries and the regions that house them.

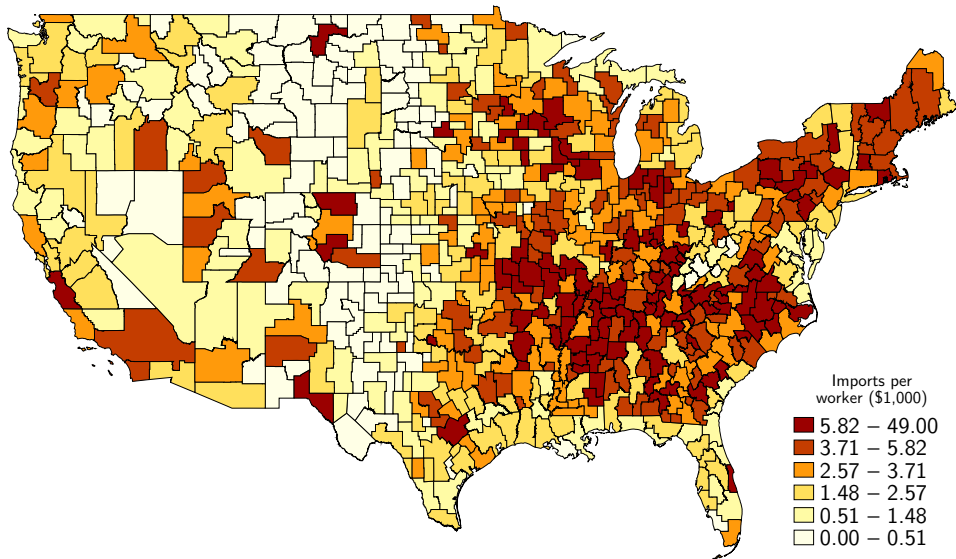
Exposed Regions

A. Exogenous exposure to robots from 1993 to 2007



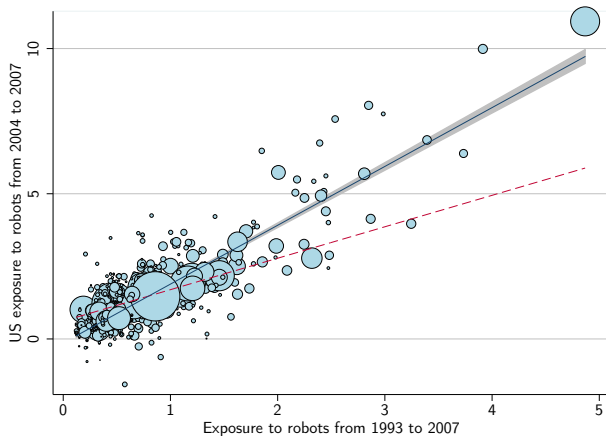
Exposed Regions

C. Exposure to Chinese imports from 1990 to 2007



First-stage Relationship

$$\text{US exposure to robots, 2004-2007}_c = \frac{2.03}{(0.27)} \cdot \text{exogenous exposure to robots, 1993-2007}_c + \hat{\nu}_c.$$



Robot-related Activities in Exposed Regions

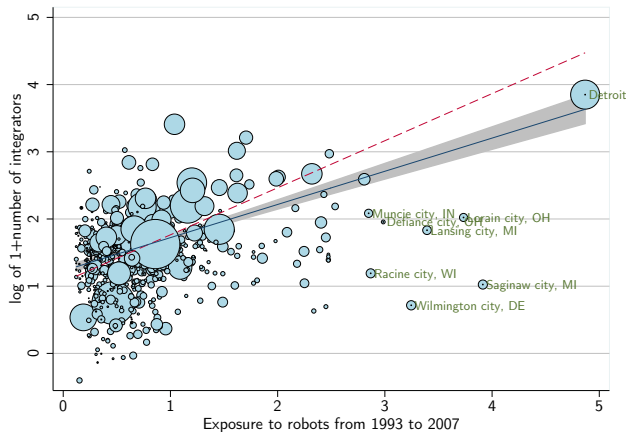


Figure: Exposure to robots predicts the location of robot integrators in the US. *Data on integrators:* Leigh and Kraft (2016)

IV Estimates for Employment

Table: IV estimates of the exposure to robots on employment

	IV ESTIMATES FROM 1990 TO 2007			
	(1)	(2)	(3)	(4)
<i>Panel A. Census private employment to population ratio.</i>				
Instrumented exposure to robots from 1993 to 2007	-0.421** (0.180)	-0.359** (0.155)	-0.367*** (0.123)	-0.371*** (0.115)
Observations	722	722	722	722
<i>Panel B. First-stage for exposure to robots in the US.</i>				
(exogenous) exposure to robots from 1993 to 2007	2.175*** (0.285)	2.178*** (0.272)	2.094*** (0.269)	2.026*** (0.275)
First-stage <i>F</i> statistic	58.3	64.2	60.6	54.2
<i>Covariates & sample restrictions:</i>				
Census division dummies	✓	✓	✓	✓
Demographics		✓	✓	✓
Broad industry shares			✓	✓
Trade, Routinization and Offshoring				✓

Regressions weighted by population. Standard errors robust against serial correlation within US states shown in parentheses.

- IV estimates: 0.37 p.p. (standard error=0.11 p.p.)

Reduced-form Impact on Employment Rate

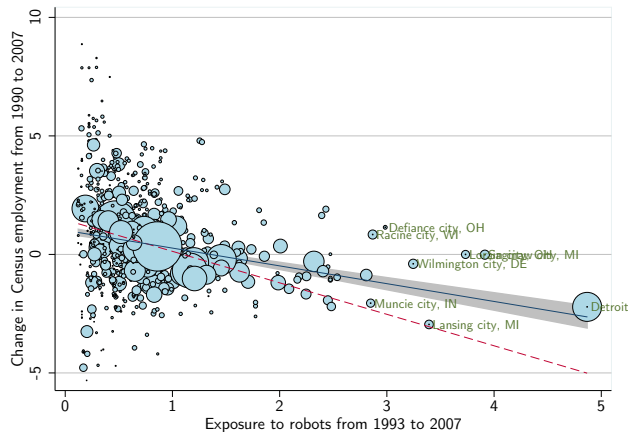


Figure: Regression plots for the impact of exposure to robots on the Census employment to population rate (in p.p.).

IV Estimates for wages

Table: IV estimates of the exposure to robots on wages

	IV ESTIMATES FROM 1990 TO 2007			
	(1)	(2)	(3)	(4)
<i>Panel A. Log of hourly wages.</i>				
Instrumented exposure to robots from 1993 to 2007	-1.043*** (0.257)	-0.891*** (0.196)	-0.675*** (0.173)	-0.732*** (0.219)
First-stage <i>F</i> statistic	60.4	66.2	61.4	55.5
Observations	163,114	163,114	163,114	163,114
<i>Covariates & sample restrictions:</i>				
Census division dummies	✓	✓	✓	✓
Demographics		✓	✓	✓
Broad industry shares			✓	✓
Trade, Routinization and Offshoring				✓

Regressions weighted by population. Standard errors robust against serial correlation within US states shown in parentheses. Model estimated within demographic×czone cells to account for observed changes in composition of wage earners.

- IV estimates: 0.75% (standard error=0.22%).
- Demographic cells: gender, age, education, race.

Reduced-form Impact on Wages

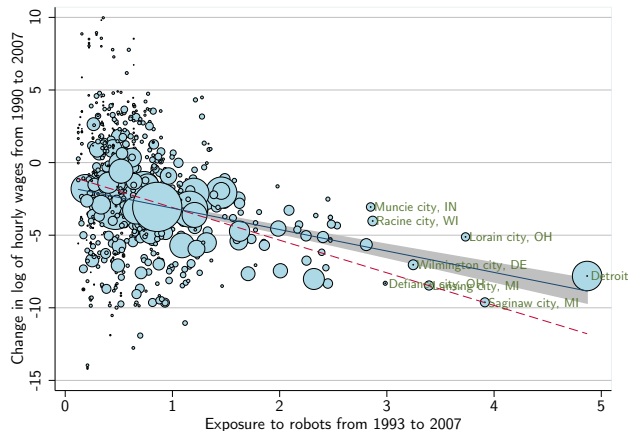


Figure: Regression plots for the impact of exposure to robots on the log of hourly wages (in log points).

Evidence Supporting Identification Assumption

- **Identifying assumption:** areas that house industries that experienced major technological developments in robotics are not on different trends.
- Show that these areas were not on a different trend in 1970-1990 (no “pre-trends”)
- Remove car manufacturing from measure of exposure.
- Stacked differences specifications where we control for czone fixed effects or shares of employment by industry.
- Compute Rotemberg weights as diagnostic tools

Placebo Exercise

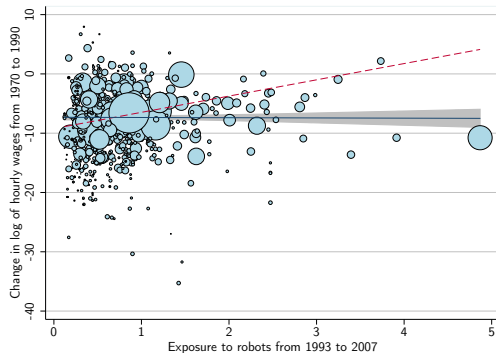
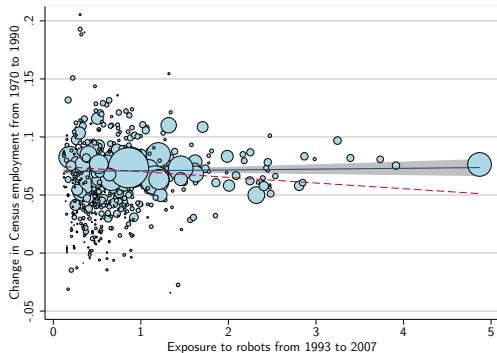


Figure: Placebo test of whether exposure to robots is correlated with past trends in employment and wages.

The Role of Car Manufacturing

Table: Estimates that control for trends in the car-manufacturing industry.

	ESTIMATES THAT CONTROL FOR CHANGES IN THE AUTO INDUSTRY					
	LONG-DIFFERENCES ESTIMATES			STACKED-DIFFERENCES ESTIMATES		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A. Census private employment to population ratio.</i>						
Exposure to robots in other industries	-1.325*** (0.363)	-1.254*** (0.342)	-1.246*** (0.283)	-0.988** (0.464)	-1.628*** (0.414)	-1.607*** (0.326)
Exposure to robots in automotive manufacture	-0.653*** (0.123)	-1.029*** (0.323)	-0.993*** (0.296)	-0.945*** (0.128)	-1.220*** (0.243)	-1.162*** (0.214)
Observations	722	722	721	1444	1444	1441
<i>Panel B. log of hourly wages.</i>						
Exposure to robots in other industries	-1.948** (0.730)	-2.269*** (0.781)	-1.846*** (0.646)	-1.800 (1.285)	-2.182** (1.000)	-0.614 (0.789)
Exposure to robots in automotive manufacture	-1.395*** (0.278)	-1.726*** (0.441)	-2.243*** (0.387)	-1.946*** (0.309)	-2.172*** (0.465)	-1.920*** (0.383)
Observations	163114	163114	160030	326377	326377	318414
<i>Covariates & sample restrictions:</i>						
Baseline covariates	✓	✓	✓	✓	✓	✓
Unweighted		✓			✓	
Down-weights outliers			✓			✓

Stacked Differences 1990-2000, 2000-2007

$$\Delta \ln Y_{tc} = \text{CZ fixed effect}_c + \gamma \cdot \text{exogenous exposure to robots}_{tc} + v_c$$

Table: Stacked-difference estimates of the exposure to robots on employment and wages

	1990-2007	Stacked differences		
	(1)	(2)	(3)	(4)
<i>Panel A. Census private employment to population ratio.</i>				
Exposure to robots	-0.751*** (0.166)	-0.951*** (0.152)	-0.953*** (0.146)	-0.613*** (0.111)
Observations	722	1444	1444	1444
<i>Panel B. log of hourly wages.</i>				
Exposure to robots	-1.476*** (0.322)	-1.939*** (0.342)	-1.919*** (0.375)	-2.519*** (0.489)
Observations	163,114	326,377	326,377	326,377
<i>Covariates & sample restrictions:</i>				
Demographic, industry shares and census division dummies	✓	✓	✓	✓
Trade, routinization and offshoring	✓		✓	✓
Commuting zone fixed effects				✓

Regressions weighted by population. Standard errors robust against serial correlation within US states shown in parentheses. Model for wages estimated within demographic×czone cells to account for observed changes in composition of wage earners.

High Quality Empirical Work

- Empirical question is challenging:
 - No natural experiment / quasi random variation
 - Low quality data on robots (only 20 industry observations)
- But the paper is convincing!
 - Enormous number of robustness tests
 - E.g. deal with automotive separately
 - Plots to “get a feel for the data”
 - Multiple tests of identification assumption
 - Different sources of variation exploited
 - E.g. stacked differences versus cross-regional
 - There are a lot more in the Appendix of the paper!

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- 2 A Model of Automation
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- 5 Estimated Effect of Robots on Jobs
- 6 Aggregate Implications of Robots**
- 7 Conclusion

Quantitative implications

- So far, estimates of the impact of robotics on exposed labor markets.
- Moving from local to aggregate effects.
- Two useful pieces of information:
 1. No evidence of out-migration.
 2. Employment losses concentrate in manufacturing, but also multiplier effects.

Adjustment of Local Labor Markets

Table: Estimates of the exposure to robots on adjustment margins

	Working-age participation		log of population		Net migration rate	
	1990-2007 (1)	Stacked (2)	1990-2007 (3)	Stacked (4)	1990-2007 (5)	Stacked (6)
Exposure to robots	-0.281*** (0.096) 722	-1.012*** (0.208) 1,444	-0.745 (0.805) 722	0.765* (0.382) 1,444	-0.031 (0.045) 722	-0.022 (0.05) 1,444
<i>Covariates:</i>						
Baseline covariates	✓	✓	✓	✓	✓	✓

Regressions weighted by population. Standard errors robust against serial correlation within US states shown in parentheses. Model for wages estimated within demographic×czone cells to account for observed changes in composition of wage earners.

- People seem to be dropping out of the labor force, but they are not moving to other industries or regions.
- Common feature of the decline in manufacturing post 1990s.

Effects on Different Industries

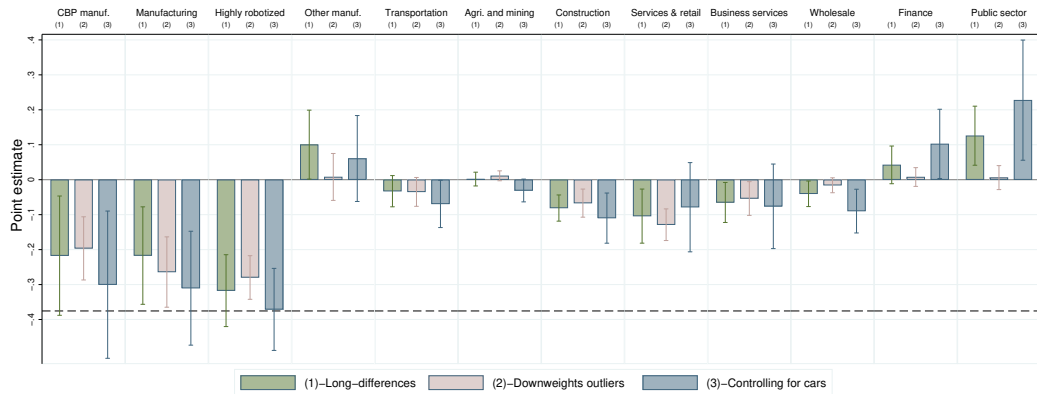


Figure: The figure plots estimates of the impact of exposure to robots on the change in employment rates separated by industry.

Aggregate Implications

- $\hat{\beta}_c^L = 0.37$: one more robot per thousand workers reduces EPOP by 0.37 p.p. in the commuting zone that adopts it *relative* to the rest.
(one robot $\rightarrow$ 6.2 jobs.)
- $\hat{\beta}_c^W = 0.73$: one more robot per thousand workers reduces wages by 0.73% in the commuting zone that adopts it *relative* to the rest.

Aggregation ignoring trade

- From 1993 to 2007, robot use in the US increased by 120,000 robots.
- 744,000 fewer jobs; average wages decline by 0.73 percent.

Aggregation Taking Trade Into Account

- Literature: $\lambda = 5$, $\sigma = 1$, $\varepsilon = 0.35$.
- Case-study evidence on the use of robots suggests that $\pi = 0.3$.
- Using IV estimates back out the key parameters η and γ :

$$\widehat{\beta}_c^L = 0.37 = \gamma (-\Phi_d^L + \Phi_\pi^L \pi) \quad \rightarrow \gamma = 6.5$$

$$\widehat{\beta}_c^W = 0.73 = (-\Phi_d^W + \Phi_\pi^W \pi) \quad \rightarrow \eta = 1.4$$

- Model implies that the aggregate effects are given by:

$$\Delta EPOP = \gamma \left(-\frac{1+\eta}{1+\varepsilon} + \frac{1+\eta}{1+\varepsilon} \pi \right) \sum_{i \in \mathcal{I}} \ell_i \frac{dQ_i}{L_i} \quad \rightarrow 0.33 \cdot \frac{dQ}{L} \text{ decline in EPOP}$$

$$\Delta \ln W = \gamma (-\eta + (1+\eta)\pi) \sum_{i \in \mathcal{I}} \ell_i \frac{dQ_i}{L_i} \quad \rightarrow 0.5 \cdot \frac{dQ}{L} \text{ percent fall in wages}$$

- 660,000 fewer jobs; average wages decline by 0.5 percent.

Some Perspective

- 660,000 jobs is a small number $\sim -0.3\%$ fall in employment-population.
No *robocalypse*.
- But industrial robots are not adopted widely yet
- US expected to add 600,000 robots in the next 10 years:
 - 2 million manufacturing jobs.
 - 4 million jobs overall.
 - 2% reduction in wages.
- These numbers miss effect of other automation technologies.
- More automation in other countries

Summarizing Acemoglu & Restrepo

- Using the case of industrial robots, Acemoglu & Restrepo argue and provide evidence that an important class of technologies can reduce employment and wages.
- Conceptually, this is possible because of the displacement of labor from automated tasks.
- Future of labor depends on types of technologies adopted.
 - Some technologies, like industrial robots, have a strong displacement effect that affects labor...
 - But other technologies may benefit it!

What About the Next Wave of Technologies?

- Next wave of technology is artificial intelligence
 - Will AI have similar effects on labor markets?
 - Affects very different workers
- This literature is being written ...

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Conclusion

- Thank you! Feedback welcome